



产品规格承认书

Product Specifications for Approval

日欣型号(NISIN Model): NS240DV4501AZ02

客户名称(Customer) : _____

客户型号(Customer Model):		
Approved by Customer		
结构(Mechanica)	电子(Electronic)	项目 (PM)

日欣光电 (NISIN Optoelectronics)		
Designed	Checked	Approved



修改记录

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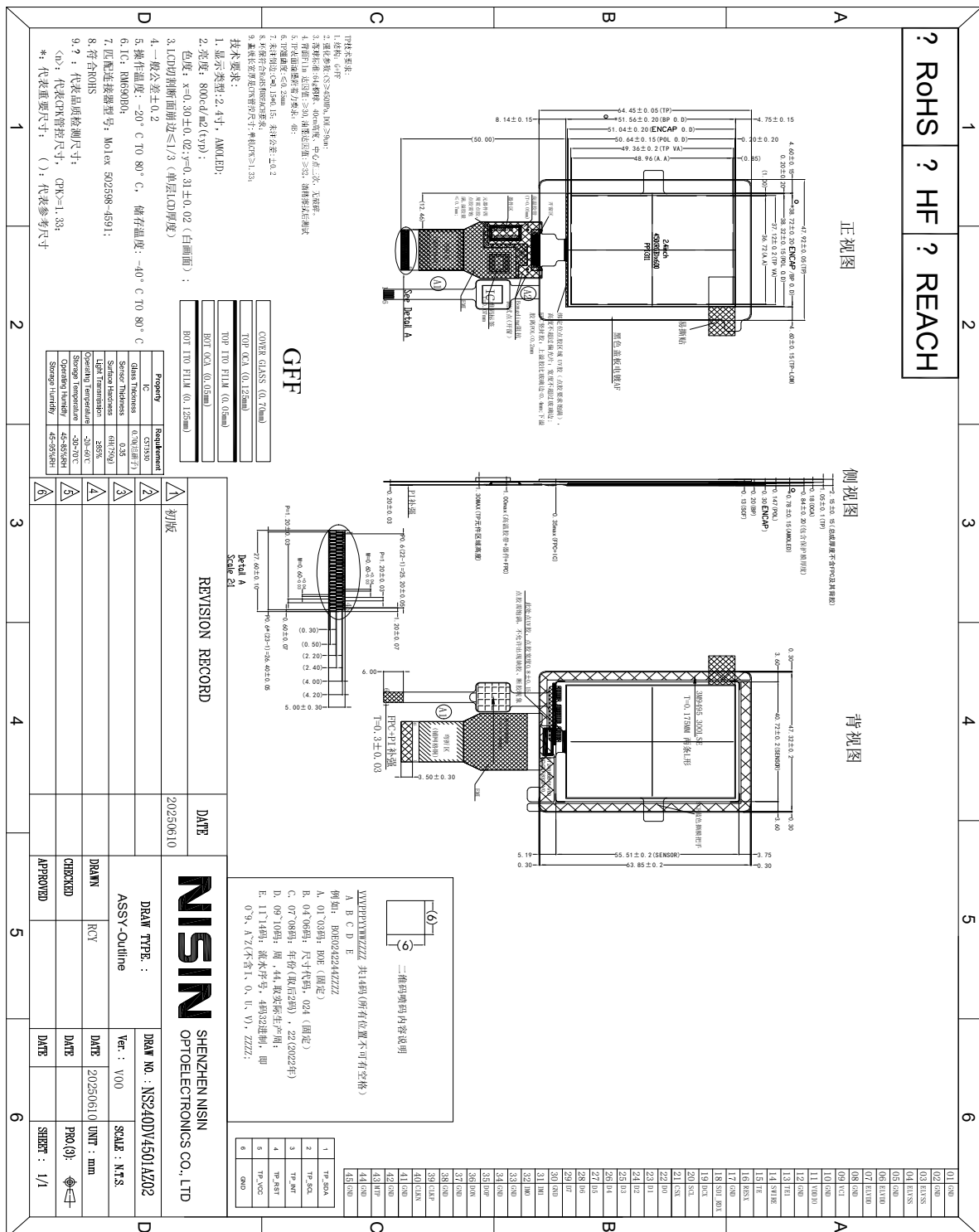
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1.产品规格（Product Specifications）

面板类型（Panel Type）	AMOLED
面板尺寸 (Panel Size)	2.4 inch
总成尺寸（Panel outline）	47.86*63.18*2.26mm
分辨率（Resolution）	450 *(RGB) * 600(dot)
显示点间距（ Dot Pitch ）	0.0816* 0.0816mm
显示色彩（color）	16.7M
视角（View Angle）	All direction
Frame rate	60Hz
Pixel Per inch	311
模组（LCM Outline）	38.72*51.56*0.78mm
Active Area	36.72*48.96mm
显示驱动 IC（Display Driver IC）	RM690B0
接口类型（Interface Type）	MIPI 1LANE/SPI/MCU
总成亮度（LCM Brightness）	Min:1200cd/m2

2.产品图纸 (Product Drawings)



3. 接口定义 (The Interface Definition)

AMOLED Pin No.	Symbol	I/O	Function
1	GND	Power	Ground.
2	GND	Power	Ground.
3	ELVSS	Power	Negative Power supply for Panel
4	ELVSS	Power	Negative Power supply for Panel
5	GND	Power	Ground.
6	ELVDD	Power	Positive Power supply for Panel
7	ELVDD	Power	Positive Power supply for Panel
8	GND	Power	Ground.
9	VCI	Power	Power supply for display driver IC analog system.
10	GND	Power	Ground.
11	VDDIO	Power	Power supply for display driver IC interface and logic system
12	GND	Power	Ground.
13	TE1	O	IC Status active reporting pin.
14	SWIRE	O	Swire protocol setting pin of Power IC
15	TE	O	Tearing effect output pin to synchronize MCU to frame writing, activated by S/W command. When this pin is not activated, this pin is output low.
16	RESX	I	Display driver reset, must be applied to properly initialize the chip. Signal is active low.
17	GND	Power	Ground.
18	SDI_RDX	I/O	SDI: Serial input signal in SPI I/F. The data is input on the rising edge of the SCL signal. RDX: Reads strobe signal to write data when RDX is "Low" in 80-series MPU interface.

19	DCX	I	Display data / command selection in 80-series MPU I/F and 4-wire SPI I/F.		
20	SCL	I	WRX : Writes strobe signal to write data when WRX is “Low” in 80-series MPU I/F. SCL: A synchronous clock signal in SPI I/F.		
21	CSX	I	Chip select input pin (“Low” enable) in 80-series MPU I/F and SPI I/F.		
22	D0	I/O	8-bit bi-directional data bus for 80-series MPU I/F and 8-bit input data bus for RGB I/F.		
23	D1	I/O			
24	D2	I/O			
25	D3	I/O			
26	D4	I/O			
27	D5	I/O			
28	D6	I/O			
29	D7	I/O			
30	GND	Power	Ground.		
31	IM1	I	IM[1:0]	Display Data	Command
			00	MIPI/3-wire SPI	MIPI/3-wire SPI
32	IM0	I	01	MIPI/4-wire SPI	MIPI/4-wire SPI
			10	MIPI/QUAD SPI	MIPI/QUAD SPI
			11	MCU 8-bit	MCU 8-bit
33	GND	Power	Ground.		
34	GND	Power	Ground.		
35	D0P	I/O	Differential data signals if MIPI interface.		
36	D0N	I/O	Differential data signals if MIPI interface.		
37	GND	Power	Ground.		
38	GND	Power	Ground.		
39	CLKP	I	Differential clock signals if MIPI interface.		
40	CLKN	I	Differential clock signals if MIPI interface.		
41	GND	Power	Ground.		
42	GND	Power	Ground.		
43	MTP	Power	MTP programming power supply. Must be left open or connected to GND in normal condition		
44	GND	Power	Ground.		
45	GND	Power	Ground.		

4.电性特性（Electrical Characteristics）

4.1. DC Characteristics Requirements

Item	Symbol	Values			Unit	Remark
		Min	Type	max		
Analog power supply Vo	VCI	2.7	3.3	3.6		
I/O Supply Voltage	VDDIO	1.65	1.8	3.3	V	
OLED input voltage	ELVDD	2.0	3.6	6	V	
OLED input voltage	ELVSS	-4.7	-3.6	-0.4	V	
Input High Voltage	VIH	0.8*VDDI	--	VDDI	V	
Input Low Voltage	VIL	0	--	0.2*VDDI	V	
Output High Voltage	VOH	0.8*VDDI	--	VDDI	V	
Output Low Voltage	VOL	0	--	0.2*VDDI	V	
Frame Frequency (60Hz)	frame		60		HZ	

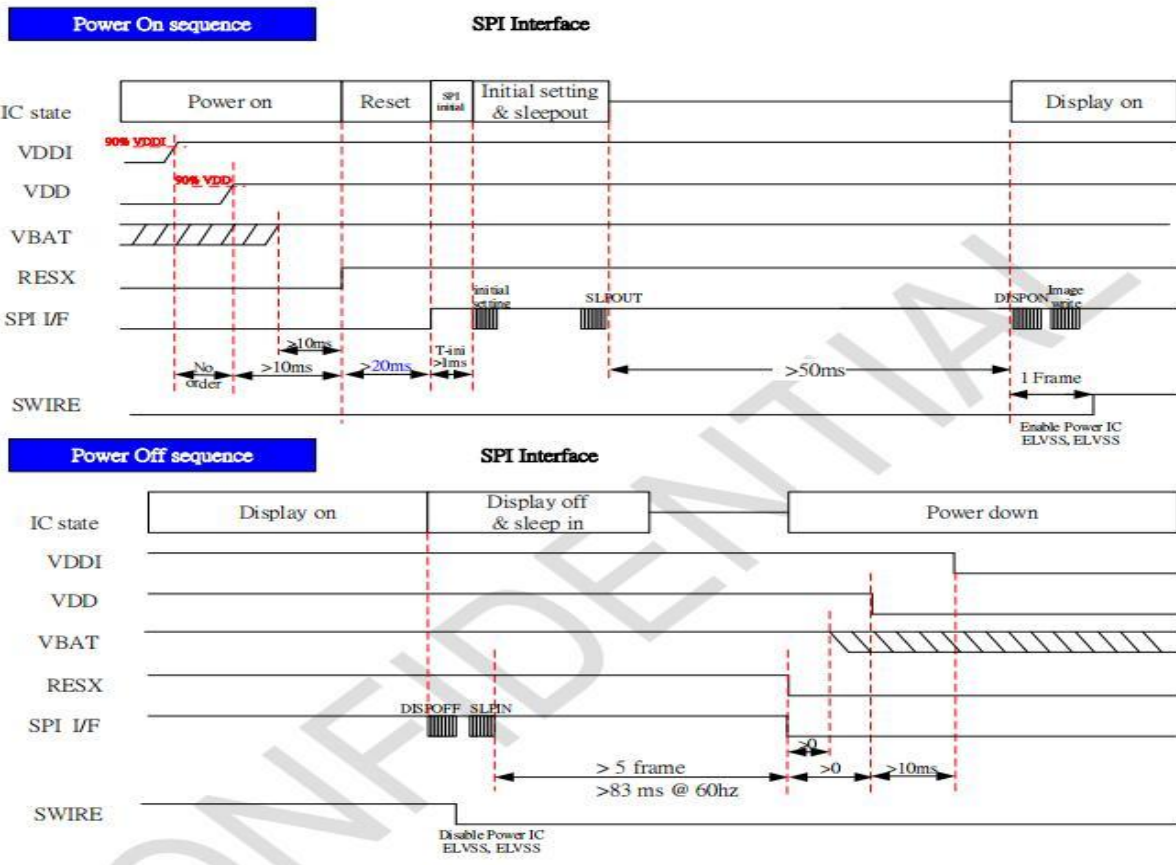
4.2. Power Consumption of Display Panel

Power Supply: VDDIO=1.8V VCI=3.3V ELVDD=3.6V ELVSS=-3.6V

Item		Symbol		Condition	Symbol	Min.	Typ.	Max.	Uni	Remark	
ELVDD		ELVDD		Normal	-	-	3.6	-	V	External Power	
ELVSS		ELVSS		Normal	-	-	-3.6	-	V		
VCI		VCI		-	-	-	3.3	-	V		
VDDIO		VDDIO		-	-	-	1.8	-	V		
Power Consumption	Display on mode (Normal)	IC	VCI	100% Pixel On,60Hz	Ivci	-	3.1	3.6	mA		
					Pvci	-	10.23	11.88	mW		
			VDDIO		Ivddio	-	1.3	1.5	mA		
					Pvddio	-	2.34	2.7	mW		
		Panel	EL		I _{ELVDD}	-	67.1	73	mA		
					I _{ELVSS}	-	67	72.4	mA		
					P _{nl}	-	483	526	mW		
Frame Rate		F _{frm}		-40℃~80℃	F _{frm}	55.2	60	64.8	Hz		
				25℃		58.2	60	61.8	Hz		

4.3. Timing Characteristics

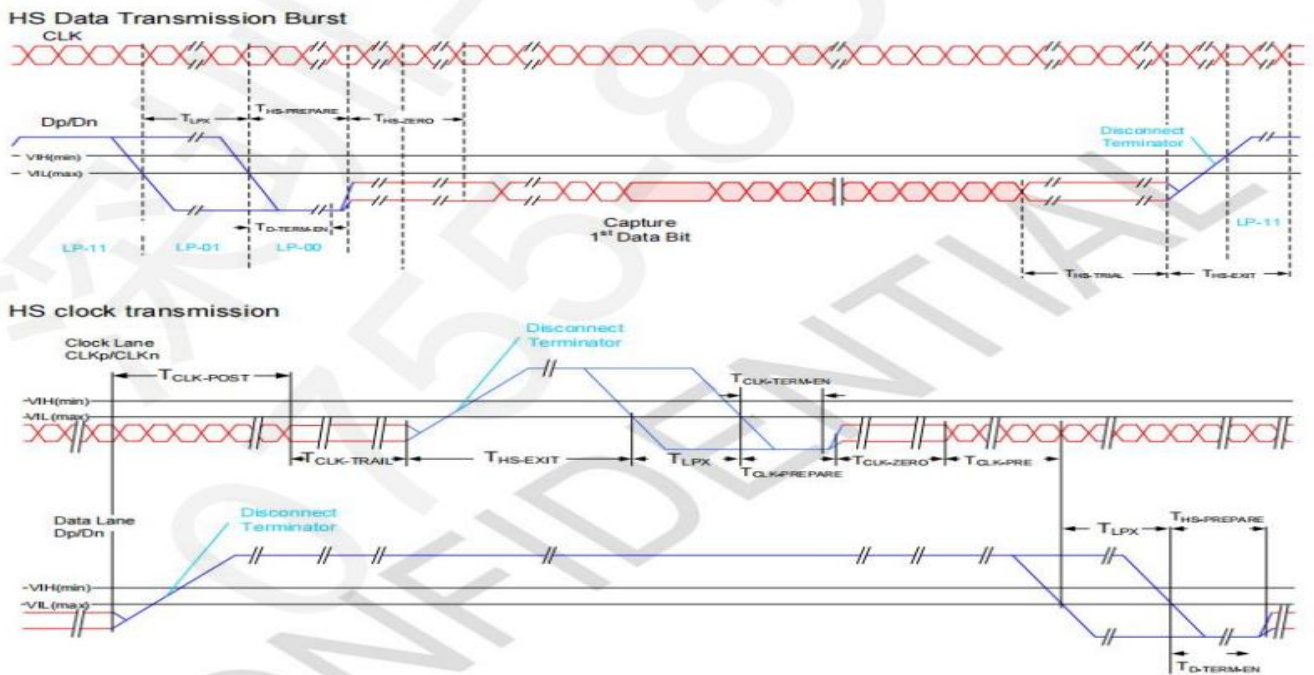
4.3.1. Display panel Power on sequence



4.3.2. Communication Interface timing

MIPI-DSI Interface & SPI&MCU Characteristics

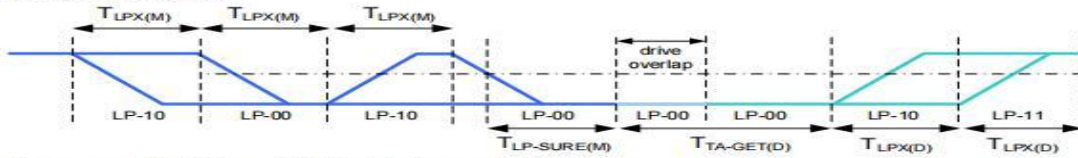
MIPI:



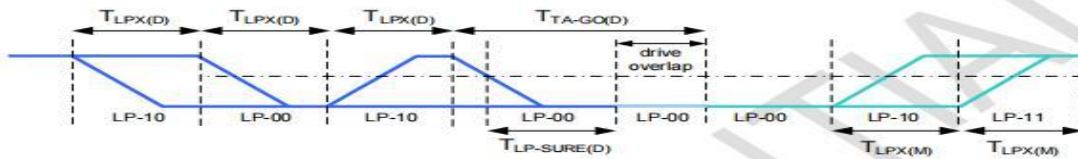
Timing Parameters:

Parameter	Description	Min	Typ	Max	Unit
$T_{CLK-POST}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to the beginning of $T_{CLK-TRAIL}$.	$60ns + 52*UI$			ns
$T_{CLK-TRAIL}$	Time that the transmitter drives the HS-0 state after the last payload clock bit of a HS transmission burst.	60			ns
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following a HS burst.	300			ns
$T_{CLK-TERM-EN}$	Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	Time for Dn to reach $V_{TERM-EN}$		38	ns
$T_{CLK-PREPARE}$	Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission.	38		95	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8			UI
$T_{CLK-PREPARE} + T_{CLK-ZERO}$	$T_{CLK-PREPARE}$ + time that the transmitter drives the HS-0 state prior to starting the Clock.	300			ns
$T_{D-TERM-EN}$	Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	Time for Dn to reach $V_{TERM-EN}$		$35 ns + 4*UI$	
$T_{HS-PREPARE}$	Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission	$40ns + 4*UI$		$85 ns + 6*UI$	ns
$T_{HS-PREPARE} + T_{HS-ZERO}$	$T_{HS-PREPARE}$ + time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.	$145ns + 10*UI$			ns
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst	$60ns + 4*UI$			ns

Turnaround Procedure



Bus turnaround (BAT) from MPU to display module timing



Bus turnaround (BAT) from display module to MPU timing

Low Power Mode :

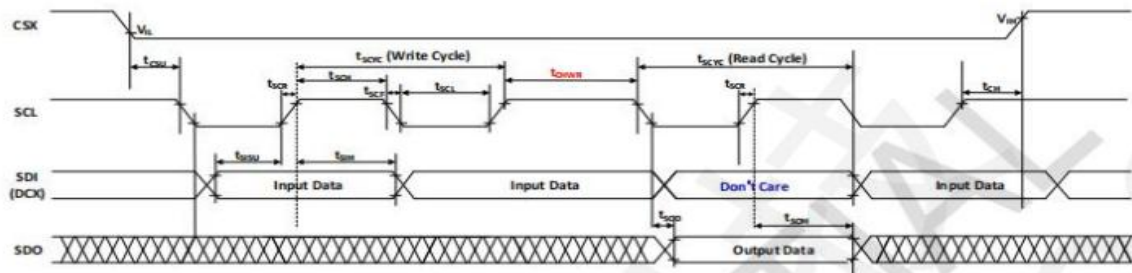
Parameter	Description	Min	Typ	Max	Unit	Notes
$T_{LPX(M)}$	Transmitted length of any Low-Power state period of MCU to display module	50		150	ns	1,2
$T_{TA-SURE(M)}$	Time that the display module waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround.	$T_{LPX(M)}$		$2*T_{LPX(M)}$	ns	2
$T_{LPX(D)}$	Transmitted length of any Low-Power state period of display module to MCU	50		150	ns	1,2
$T_{TA-GET(D)}$	Time that the display module drives the Bridge state (LP-00) after accepting control during a Link Turnaround.		$5*T_{LPX(D)}$		ns	2
$T_{TA-GO(D)}$	Time that the display module drives the Bridge state (LP-00) before releasing control during a Link Turnaround.		$4*T_{LPX(D)}$		ns	2
$T_{TA-SURE(D)}$	Time that the MPU waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround.	$T_{LPX(D)}$		$2*T_{LPX(D)}$	ns	2

NOTE:

1. T_{LPX} is an internal state machine timing reference. Externally measured values may differ slightly from the specified values due to asymmetrical rise and fall times.
2. Transmitter-specific parameter

SPI:

3/4-wire SPI



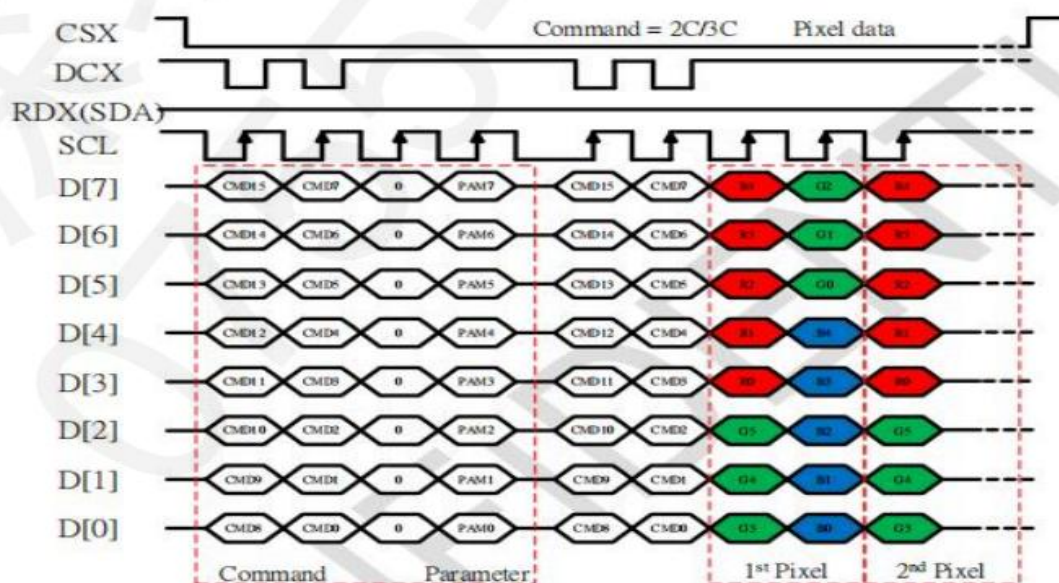
Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Clock cycle	t _{SCYC}	Write	20			ns
		Read	300			ns
Clock high pulse width	t _{SCH}	Write	6.5			ns
	t _{SCH}	Read	140			ns
Clock low pulse width	t _{SCL}	Write	6.5			ns
	t _{SCL}	Read	140			ns
Clock rise time	t _{SCR}	0.2*VDDI -> 0.8*VDDI			3.5	ns
Clock fall time	t _{SCF}	0.8*VDDI -> 0.2*VDDI			3.5	ns
Chip select setup time	t _{CSU}		10			ns
Chip select hold time	t _{CH}		10			ns
Data input setup time	t _{SISU}	To V _{IL} of SCL's rising edge	5			ns
Data input hold time	t _{SIH}		5			ns
Access time of output data	t _{SOD}	From V _{IL} of SCL's falling edge			120	ns
Hold time of output data	t _{SOH}	From V _{IH} of SCL's rising edge	5			ns
Transition time from Write cycle to Read cycle	t _{CHWR}	From V _{IH} of SCL's rising edge	150			ns

Notes:

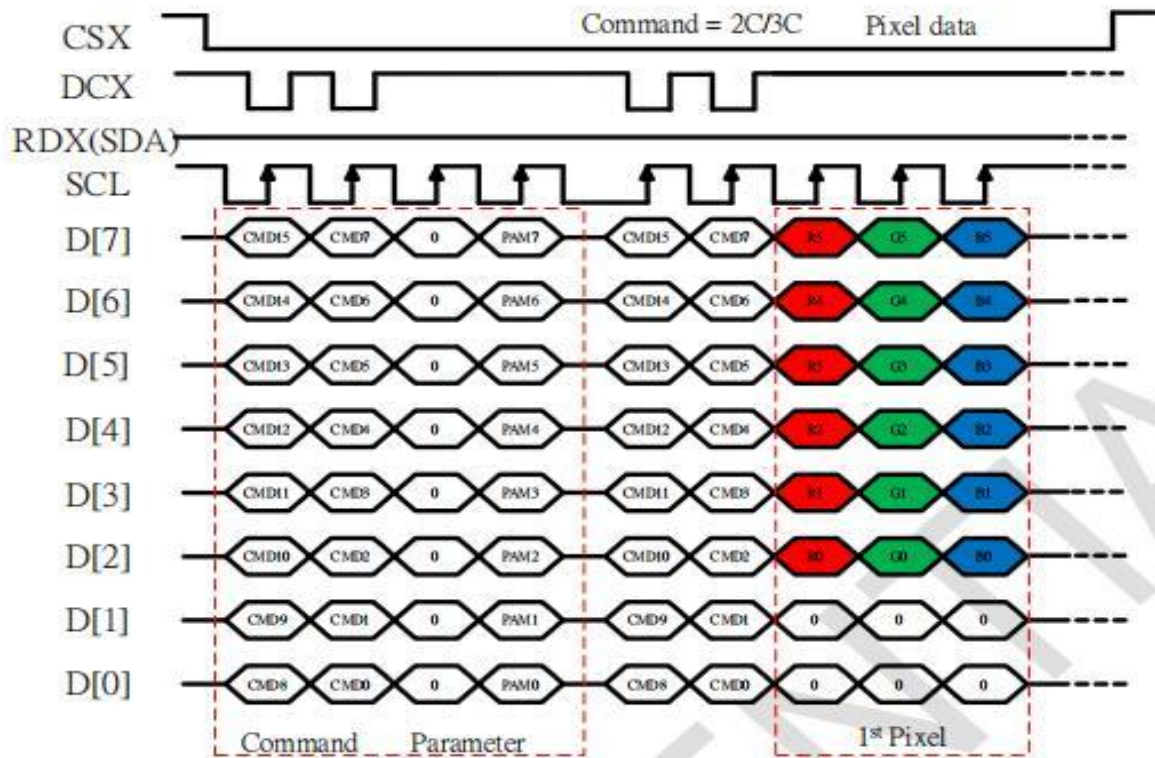
- (1) Logic high and low levels are specified as 80% and 20% of VDDI for Input signals.
- (2) For the 4-wire SPI, the DCX's timing is the same as input data.
- (3) $T_a = -30^{\circ}\text{C}$ to 70°C , VDDI=1.65V to 3.3V, VDD=2.7V to 3.6V, and VSS=0V

MCU:

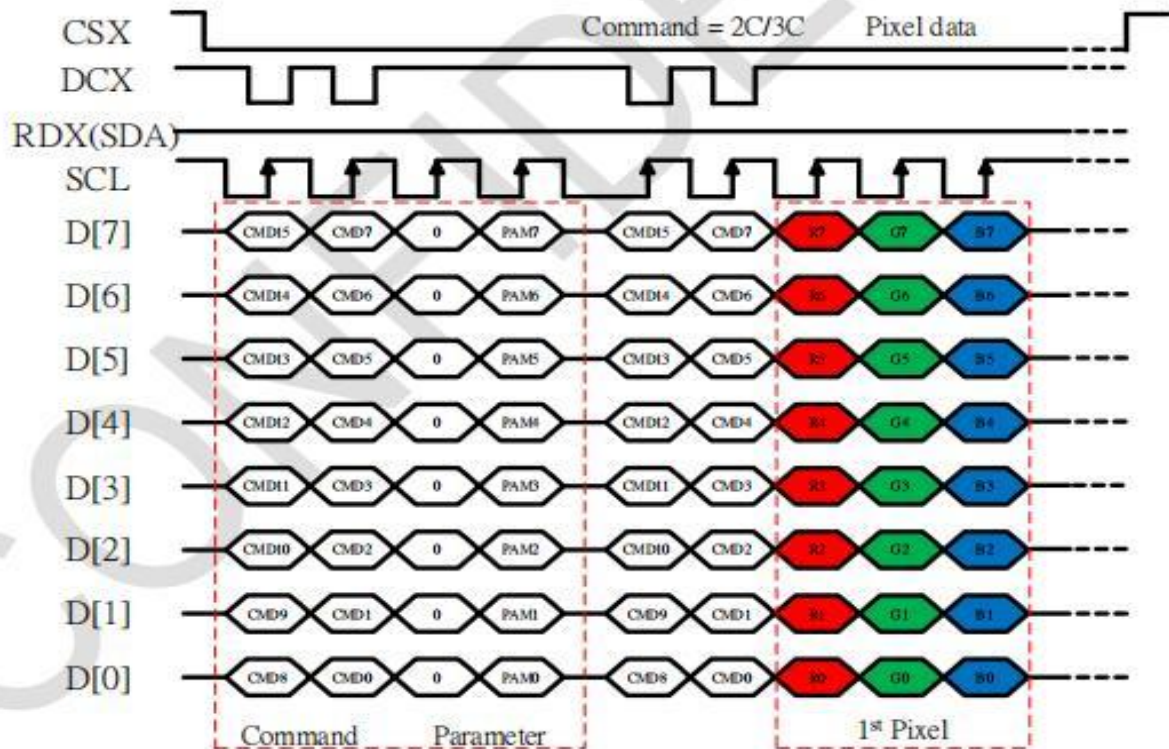
The basic command format and IFPF (0x3A00, interface pixel format) is equal to 5:



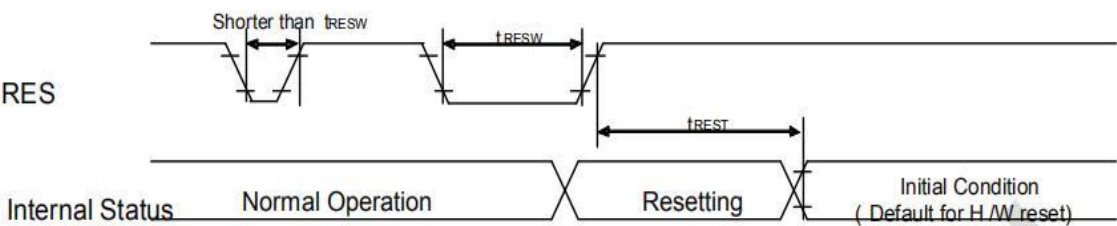
The basic command format and IFPF (0x3A00, interface pixel format) is equal to 6:



The basic command format and IFPF (0x3A00, interface pixel format) is equal to 7:



4.3.3. Display panel reset timing:

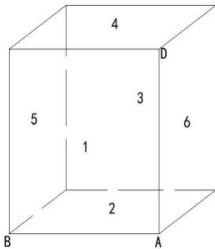


Reset input timing:
VDDI=1.65 to 3.3V, VDD=2.7 to 3.6V, AGND=DGND=0V, Ta=-40 to 85°C

Symbol	Parameter	Related Pins	MIN	TYP	MAX	Note	Unit
t_{RESW}	*1) Reset low pulse width	RESX	30	-	-	-	μs
t_{REST}	*2) Reset complete time	-	-	-	20	When reset applied during Sleep in mode	ms
		-	-	-	120	When reset applied during Sleep out mode	ms

5.可靠性实验测试(Reliability Test Conditions And Methods)

序号	试验项目	试验条件及方法	试验设备	检验项目	检验工具
1	高温高湿(静、动态)试验	温度 $60^{\circ}\text{C} \pm 3^{\circ}\text{C}$, 湿度 $90\% \pm 3\%$, 要求选择时间分别为 96 小时, 静、动态(产品点亮)在室温下恢复 2 小时后进行外观, 显示功能全画面检查。	恒温恒湿试验机	检验外观、功能、抗腐蚀性	目视/测试架/客户样机/显微镜
2	高、低温冲击试验	静态 -30°C (30 分钟) $\sim 80^{\circ}\text{C}$ (30 分钟) $\sim -30^{\circ}\text{C}$ (30 分钟), 24 个循环, 在室温下恢复 2 小时后进行外观, 显示功能全画面检查。	冷热冲击试验机	检验外观、功能	
3	高温贮存试验	常温 $70^{\circ}\text{C} \pm 3^{\circ}\text{C}$ 、宽温 $80^{\circ}\text{C} \pm 3^{\circ}\text{C}$ 、96 小时后在室温状态下恢复 1 小时在 2 小时内完成外观、显示功能全画面检查。	烤箱	检验外观、功能	目视/测试架/客户样机
4	低温贮存试验	常温 $-20^{\circ}\text{C} \pm 3^{\circ}\text{C}$ 、宽温 $-30^{\circ}\text{C} \pm 3^{\circ}\text{C}$ 、条件的试验箱内保存 96 小时后在室温状态下恢复 1 小时, 在 2 小时完成外观、显示功能全画面检查, 特别注意检查是否有漏液、断线、腐蚀、偏光片不良现象。	低温冰箱	检验外观、功能	
5	低温贮存试验(动态)	常温 $-20^{\circ}\text{C} \pm 3^{\circ}\text{C}$ 、宽温 $-30^{\circ}\text{C} \pm 3^{\circ}\text{C}$ 条件的试验箱内点亮刷屏, 过程中每 1 小时观察一次, 全画面检查显示功能如: 异常, 卡机, 花屏等。特别注意检查是否有漏液、断线、腐蚀、偏光片不良现象。	低温冰箱	检验外观、功能	目视/测试架/客户样机

6	包装模组跌落试验	1、跌落重量及自由落体高度： (图二) 2、自由落体角度如下： <table><tr><th>总重量</th><th>自由落体高度</th></tr><tr><td>0-9kg</td><td>92cm</td></tr><tr><td>9-25kg</td><td>76cm</td></tr><tr><td>25-45kg</td><td>53cm</td></tr><tr><td>45-68kg</td><td>46cm</td></tr><tr><td>大于 68kg</td><td>41cm</td></tr><tr><td></td><td></td></tr></table> 5, 面 6;  1) 一角: A 角 2) 三菱: A-B, A-D, A-C 3) 六面: 面 1, 面 2, 面 3, 面 4, 面 5, 面 6;	总重量	自由落体高度	0-9kg	92cm	9-25kg	76cm	25-45kg	53cm	45-68kg	46cm	大于 68kg	41cm			包装模组跌落架	测试电性能无异常、外观检验无破损，无脱离现象	目视/测试架/客户样机
总重量	自由落体高度																		
0-9kg	92cm																		
9-25kg	76cm																		
25-45kg	53cm																		
45-68kg	46cm																		
大于 68kg	41cm																		
7	盐雾试验	标准条件:中性盐雾试验 (NSS 试验): 5%的氯化钠盐水溶液, 溶液 PH 值中性 (6.5~7.2), 试验温度 $35 \pm 2^{\circ}\text{C}$, 盐雾的沉降率在 $1 \sim 2\text{ml}/80\text{cm}^2 \cdot \text{h}$ 之间, 时间 24h。2. 其它特殊要求条件:醋酸盐雾试验 (ASS 试验): 5%氯化钠溶液中配入冰醋酸, 溶液 PH 值为 3 左右, 试验温度 $35 \pm 2^{\circ}\text{C}$, 盐雾的沉降率在 $1 \sim 2\text{ml}/80\text{cm}^2 \cdot \text{h}$ 之间, 时间 24h。全画面检查外观、功能, 特别注意检查是否有腐蚀	盐雾试验设备	检验外观、功能, 盐雾试验结果的判定方法, 腐蚀物出现判定法: 定性判定, 试验后功能测试应 OK, 外观观察产品无腐蚀现象产生。	目视/测试架/客户样机/显微镜														
8	ESD 抗静电试验	1)装整机状态下试验:接触 4KV, 非接触 8KV 放电测试, 接触测试时, LCM 通电状态下测 5 个点。 2) 测试架测试状态下试验: 接触 4KV, 非接触 8KV 放电测试, 接触测试时, LCM(或总成)通电状态下测 9 个点。其中产品正面显示区域两个点, 产品显示区域背面 4 个点, 产品左侧面、右侧面和 IC 端对面中间位置各一个点。(如 6.17.3.3 图所示) 测试点 1—6 是空气放电, 测试点 7—9 是接触放电, 10 次/点。总成(或模组)实验后在图案、黑色、白色、灰阶画面各撕膜 3~5 次, 确认有无撕膜静电不良, 撕膜静电速度适中, 不可过快及刻意放慢。判定标准: 撕膜 3 次确认无撕膜静电 , OK。撕膜 3 次以上出现撕膜静电, 在保护膜覆盖后 3 秒内静电消失 OK。撕膜出现撕膜静电, 在保护膜覆盖后 3 秒内静电不消失 NG。	抗静电枪 (尖头接触放电, 圆头空气放电)	检验外观、功能	目视/测试架														

6. 光电参数 (Optical Characteristics)

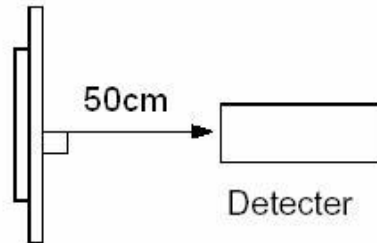
Test condition: VCI= 3.3V, VDDIO=1.8V , ELVDD=3.6V ,ELVSS=-3.6V,Ta=25°C

Item	Symbol	Condition	Min	Typ	Max	Unit	Note
Luminance (with pol&lens)	Normal mode Bp	$\theta=0^\circ \phi=0^\circ$	720	800	880	cd/m2	
Uniformity	ΔBp		90	--	--	%	
Viewing Angle	Left	θL	80	--	--	deg	
	Right	θR	80	--	--		
	Top	ψT	80	--	--		
	Bottom	ψB	80	--	--		
Contrast Ratio	Cr	$\theta=0^\circ \phi=0^\circ$	10000 00: 1		--	-	
Response Time	Tr		--	--	2	ms	
	Tf		--	--	2	ms	
	Tgray		--	--	2	ms	
Color Coordin ate of CIE193 1	Red	x	0.660	0.680	0.700		
		y	0.300	0.320	0.340		
	Green	x	0.205	0.245	0.285		
		y	0.675	0.715	0.755		
	Blue	x	0.121	0.141	0.161		
		y	0.023	0.043	0.063		
	White	x	0.280	0.300	0.320		
		y	0.290	0.310	0.330		
Color temperature	CT		7000	7500	8000	K	
Flicker	amount	60HZ @Interval screen(间隔画面)	--	--	-30	dB	
	amount	60HZ @255Gray	--	--	-50	dB	
Gamma	Normal mode		2.0	2.2	2.4		
Polarization direction of front polarizer	PdF		--	135	--	deg	

Color shift		$\theta L=30^\circ$	--	3	4	JNCD	
		$\theta R=30^\circ$	--	3	4	JNCD	
		$\psi T=30^\circ$	--	3	4	JNCD	
		$\psi B=30^\circ$	--	3	4	JNCD	
OLED lifetime	LT95	At 25°C,with white color pattern	290	--	--	hrs.	
Color gamut	DCI-P3	Ratio	97	100		%	

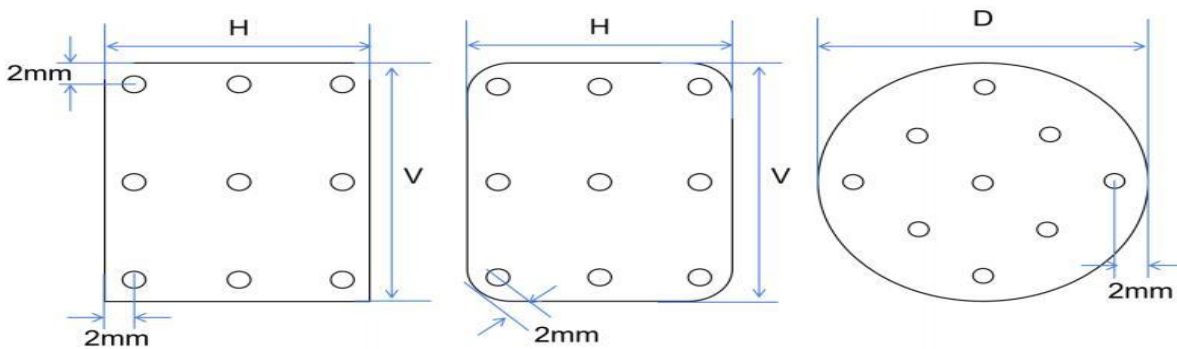
6.1. Luminance measurement:

The test condition is at 25°C and measured on the surface of OLED module. • The data are measured after OLEDs are lighted on for more than 5 minutes and displays • equipment CS2000/CS2000A or similar equipments (Field of view:1deg,Distance:50cm) • Measuring surroundings: Dark room. • Adjust operating voltage to get optimum contrast at the center of the display. • Measured value at the center point of panel must be after more than 5 minutes while light up



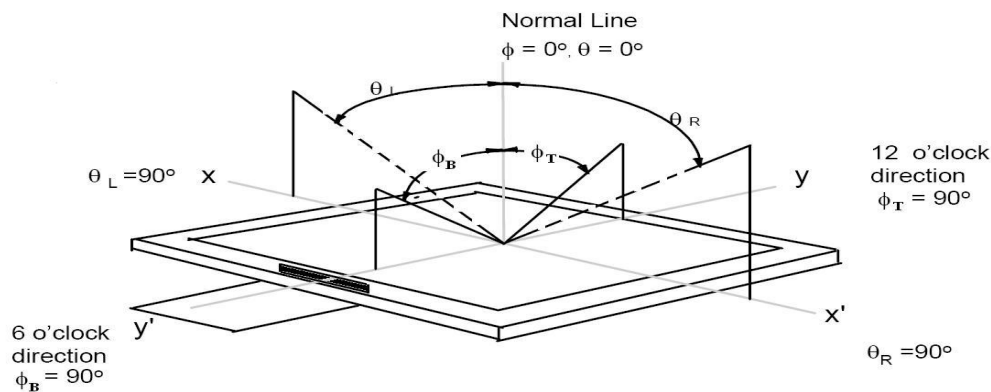
6.2. Uniformity:

- The test condition is at 25°C and measured on the surface of display module
- Measurement equipment: CS2000/CS2000A or similar equipment.
- The luminance uniformity is calculated by using following formula. $\Delta Bp = Bp (\text{Min.}) / Bp (\text{Max.}) \times 100 (\%)$ • $Bp (\text{Max.})$ = Maximum brightness in 9 measured spots
- $Bp (\text{Min.})$ = Minimum brightness in 9 measured spots.



6.3: The definition of Viewing Angle

Refer to the graph below marked by θ and Φ



6.4: The definition of Contrast Ratio (Test OLED using CS2000/CS2000A or similar equipments):

Contrast Ratio(CR)=

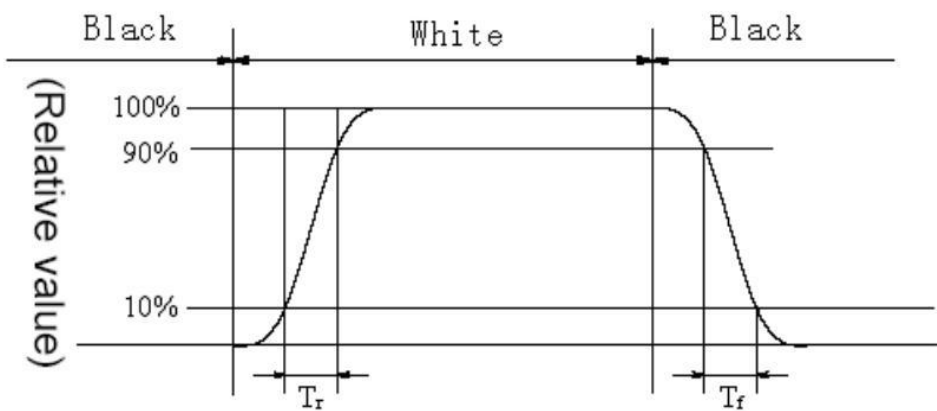
Luminance is at "White" state

Luminance is at "Black" state

(Contrast Ratio is measured in optimum common electrode voltage. Black state display pure black color and luminance<0.002nits.)

6.5: Definition of Response time. (Test module using DMS 803 or similar equipment):

The output signals of photo detector are measured when the input signals are changed from "black" to "white"(Voltage falling time) and from "white" to "black"(Voltage rising time), respectively. The response time is defined as the time interval between the 10% and 80% of amplitudes. Refer to figure as below.



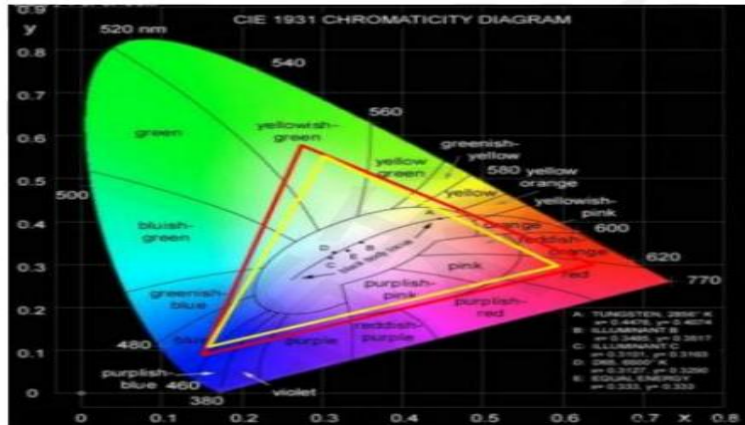
Response time of gray to gray:

- Measurement equipment: DMS803 or similar equipment.
- Test method: we define 8 grays L0-L7, the grays of L0-L7 were defined as:0,36,73, 109, 146, 182, 219, 255. The output signals of photo detector are measured when the input signals are changed from "Lx" to "Ly", x, y= [0, 7]. The response time is defined as the time interval between the 10% and 90% of amplitudes. The result of the test can be noted as below:

	L0	L1	L2	L3	L4	L5	L6	L7
L0								
L1								
L2								
L3								
L4								
L5								
L6								
L7								

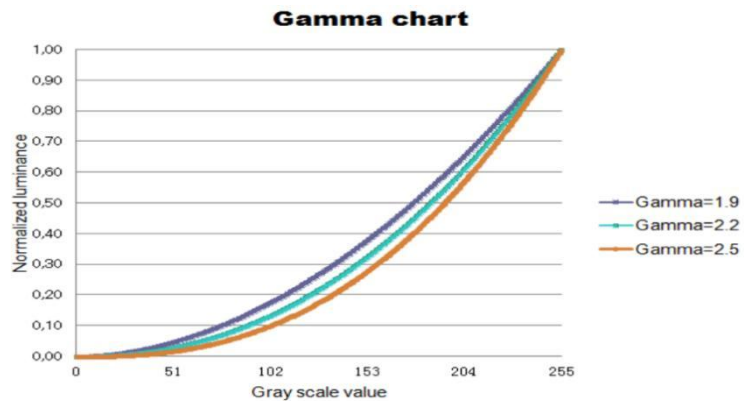
6.6: Definition of Color of CIE Coordinate and DCI-P3 Ratio.

$$S = \frac{\text{area of RGB triangle}}{\text{area of DCI-P3 triangle}} \times 100\%$$



6.7: gamma curve control

- For gamma curve control, request as below:
- the whole curve's tolerance must control within +/-0.3, will test the gray scale below: 0~255gray, interval 8 gray (0,8,16,24.....255)
- According to below formula chart the curves



$$TR\%x = \sqrt{\frac{Gx - G0}{G255 - G0}}$$

3.10 OLED lifetime

Test samples 5pcs;

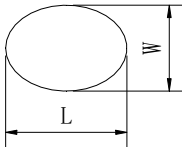
Measurement equipment: CS2000/CS2000A or similar equipment.

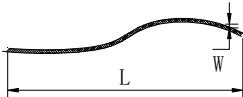
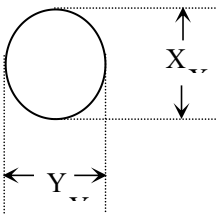
At room temperature(25℃), light the module with typical value brightness, display a white pattern, To record the brightness every 24 hours.

LT95>290h.

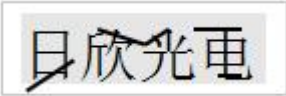
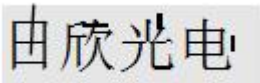
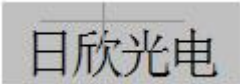
7.检验标准 (Inspection standard)

7.1、外观检验标准：

项目	不良描述	判定标准			检验方法
1-1 外观 (次缺)	产品外观或点亮状态下的圆形物 (LCD/POL/BL/盖板/TP/组装异物/贴合异物、产品本身的亮点/暗点等)  Φ=(L+W)/2	> 2.4 寸——7.0 寸			目视 (菲林卡比对)
		点状直径	允许个数	备注	
		Φ≤0.10mm	不计	1: 10mm 间距内只允许 3 个点。 2: 显示区只允许 10 个点，超过以上任意一项则 NG	
		0.1mm < Φ≤0.15mm	4	TP、屏各允许 2 个	
		0.15mm < Φ≤0.2mm	2	TP、屏各允许 1 个	
		Φ>0.2mm	0		

1-2 外观 (次 缺)	产品外观或点亮状态下的 的线状物 (LCD/POL/BL/ 盖板/TP/组装异物/贴合 异物、产品本身的亮点/ 暗点等) 	<table><tr><td>长 (L)</td><td>宽 (W)</td><td>允许 个数</td></tr><tr><td>≤1mm</td><td>≤0.03mm</td><td>2</td></tr><tr><td>≤2mm</td><td>0.03<W≤0.05mm</td><td>1</td></tr><tr><td>>2mm</td><td>>0.05mm</td><td>NG</td></tr></table> 注意：两条线之间必须距离 10mm 以上（>2.4 寸—7.0 寸）。	长 (L)	宽 (W)	允许 个数	≤1mm	≤0.03mm	2	≤2mm	0.03<W≤0.05mm	1	>2mm	>0.05mm	NG	目视 (菲 林卡 比对)
长 (L)	宽 (W)	允许 个数													
≤1mm	≤0.03mm	2													
≤2mm	0.03<W≤0.05mm	1													
>2mm	>0.05mm	NG													
1-3 外观 (次 缺)	气泡： $\Phi = (X+Y) / 2$ 	<table><tr><td>尺寸</td><td>允许个数</td></tr><tr><td>1、$\Phi \leq 0.1\text{mm}$ 2、不超过边框 1/3</td><td>不计 (密集不可)</td></tr><tr><td>$0.10 < \Phi \leq 0.2\text{mm}$</td><td>1</td></tr><tr><td>$\Phi > 0.2\text{mm}$</td><td>NG</td></tr><tr><td>$0.2 < \Phi \leq 1.5\text{mm}$, (边框以外)</td><td>3</td></tr></table> 注意：>2.4 寸-6.0 寸气泡间距大于 10mm 以上	尺寸	允许个数	1、 $\Phi \leq 0.1\text{mm}$ 2、不超过边框 1/3	不计 (密集不可)	$0.10 < \Phi \leq 0.2\text{mm}$	1	$\Phi > 0.2\text{mm}$	NG	$0.2 < \Phi \leq 1.5\text{mm}$, (边框以外)	3	在日 光台 灯下 撕起 保护 膜, 距 待测 物 30cm 目视		
尺寸	允许个数														
1、 $\Phi \leq 0.1\text{mm}$ 2、不超过边框 1/3	不计 (密集不可)														
$0.10 < \Phi \leq 0.2\text{mm}$	1														
$\Phi > 0.2\text{mm}$	NG														
$0.2 < \Phi \leq 1.5\text{mm}$, (边框以外)	3														
1-4 外观 (次 缺)	导电布、胶带贴附不良	1、ITO 点银浆位置贴附导电布贴偏超出 LCD 边缘 NG 2、导电布未连接上玻璃 NG 3、导电布贴附位置与图纸不符 NG 4、漏贴导电布、IC 防紫外线胶、FPC 保护胶 NG	目视												

1-5 外观 (次缺)	焊接不良 1、A/K 点或 T/P 焊接点歪、锡量超厚、未焊接、虚焊、假焊。 2、A/K 点或 T/P 焊接点未清洁。 3、焊接线沾锡 4、T/P 焊接对位	1、A/K 焊点按搭接面积大于 2/3 OK，焊点厚度不超过工程图上限，A/K 或 TP 焊点未焊接、虚焊、假焊 NG 2、A/K 点或 T/P 焊接点有残渣、松香油、脏污等 NG 3、FPC 焊接线有锡，分开包装，提示入库。 4、以焊接丝印线为基准，下焊超过丝印 0.2mm NG。 上焊与丝印靠齐距离大于 0.5mm NG.	目视 (必要时放大镜检查确认)
1-6 外观 (次缺)	FPC 弯折 1、FPC 边缘未与 BL 边缘平齐，压印线未与定位线对准。 2、元器件顶住背光胶框。	1. FPC 反折要用反折治具检查，未折到位，FPC 边缘须与 BL 边缘平齐，压印线定位孔未对准， NG； 2. 用手指摸背光背面元件区域平整，目视元件有无顶住 BL 边缘，并且预留 0.2MM 间距 OK	卡尺 测量
7.2、电讯检验标准：			
项目	不良描述	判定标准	检验方法
2-1 电讯 (主缺)	漏液晶	LCD 因密封性受损导致空气进入液晶盒内 NG	目视，必要时投低温验证
2-2 电讯 (主缺)	少划： LCD 点亮后有整行(或列)或字段点阵不显的现象	日欣光电 NG 日欣光电 OK	用测架点亮 LCD 后目视

2-3 电讯 (主 缺)	乱讯：在显示字体画面 时有不该显示的地方显 示(画面乱七八糟)	 NG	用测 架点 亮 LCD 后目 视
2-4 电讯 (主 缺)	字节不均(暗画)：需要 显示的字节本身有明暗 不一致现象	 NG	用测 架点 亮 LCD 后目 视
2-5 电讯 (主 缺)	多画：有不需显示之 字节出现在显示区内	 NG	用测 架点 亮 LCD 后目 视
2-6 电讯 (主 缺)	显示深浅：比对样品有 字深字淡现象(在测试 条件一致的情况下量测 产品 Vop 值是否在偏差 范围内)	 深 标准 浅	必要 时可 用主 机板 确认
2-7 电讯 (主 缺)	无显：通电后 LCD 无动 作或无画面	NG	用测 架点 亮 LCD 后目 视

8.模组使用注意事项（Precautions for Use of LCD Modules）

8.1 如果接口定义内有定义 IM0，请根据规格书（4.接口定义）内的定义做正确选择以匹配数据线的位数。

8.2 客户在做结构设计时，请保证机壳开窗尺寸比触摸屏 V.A 单边少 0.3mm。泡棉开窗尺寸比触摸屏 V.A 单边大 0.2mm。

8.3 模组的主要部件 LCD 和 TP 都是由玻璃组成，在测试、使用、移动过程中，请轻拿轻放。当产品不带触摸屏时，靠近 FPC 的屏幕两端绝对不能受力，否则会导致玻璃破损和显示不良的发生。

8.4 粘合偏光片、背光、触摸屏的胶材是有机物质，在接触到甲苯、乙醇、丙酮时，会破坏粘性。在使用中，请防止这些物质接触到产品。

8.5 如果显示表面掉落有灰尘、异物，切忌用手直接擦拭。请用棉签轻轻挑擦。

8.6 如果 LCD 破损导致液晶泄露，请不要让皮肤或衣服沾到液晶。如果不小心碰到，请立即用肥皂和清水清洗。

8.7 用手直接触摸显示区域会造成偏光片的损坏，同时容易引起静电问题。

8.8 当模组运行时，在显示区域施加压力会导致显示不正常。撤去外力，重新开机，可以恢复。

8.9 潮湿的环境可能引起玻璃 ITO 的腐蚀，在使用中，在保证规格书要求的最高温度条件下请确保湿度在 60%以下。